Personal Expenses Tracker

A

Major Project

Submitted in partial fulfilment for the award of the degree of

BACHELORS OF TECHNOLOGY

IN

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GWALIOR • MP • INDIA

by

Priyanshu Rathore (BCAN1CA23068)

Gautam Yadav (BCAN1CA23031)

Nikhil Kushwaha(BCAN1CA23053)

Under the guidance of

Ms.Anjali Saraswat

Department of Computer Science Engineering

School of Engineering and Technology

ITM UNIVERSITY, GWALIOR - 474006 MP, INDIA

(April 2025)

CERTIFICATE

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We further declare that the work is free from any plagiarism.

(Signature of the candidates)

(Priyanshu Rathore)

(Gautam Yadav)

(Nikhil Kushwah)

(BCAN1CA23068)

(BCAN1CA23031)

(BCAN1CA23053)

Verified by guide

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Abstract

In today’s digital age, managing personal finances efficiently is essential for maintaining financial stability. Many individuals find it difficult to keep track of their daily income and expenses, which often leads to poor budgeting and unnecessary spending. To overcome this problem, the “Personal Finance Management System” has been developed as a simple and user-friendly web application.

The system allows users to record their income and expenses by entering transaction details such as description and amount. It automatically calculates the total balance and provides real-time updates, helping users maintain a clear understanding of their financial status. The application also categorizes transactions into income and expenses, making it easier to analyze financial patterns.

A key feature of the system is the use of graphical representation through charts, which visually displays the monthly summary of income and expenses. This helps users quickly understand their financial behavior and make better decisions.

The project is developed using HTML, CSS, and JavaScript, and uses localStorage for data persistence, eliminating the need for an external database. This makes the system lightweight, fast, and easy to use.

In conclusion, the Personal Finance Management System is an effective tool for managing daily financial activities and demonstrates the practical use of web technologies in solving real-world problems.

Content

List of Abbreviations

CSE – Computer Science Engineering

HOD – Head of department

HTML – HyperText Markup Language

CSS – Cascading Style Sheets

JS – JavaScript

UI – User Interface

UX – User Experience

PFMS – Personal Finance Management System

DB – Database

API – Application Programming Interface

OS – Operating System

RAM – Random Access Memory

CPU – Central Processing Unit

URL – Uniform Resource Locator

HTTP – HyperText Transfer Protocol

HTTPS – HyperText Transfer Protocol Secure

JSON – JavaScript Object Notation

DOM – Document Object Model

LS – Local Storage

LIST OF FIGURES & TABLES

CHAPTER 1: INTRODUCTION & OBJECTIVES

1.1 Introduction

In today's digital world, managing personal finances has become an essential task. Many individuals find it difficult to keep track of their daily income and expenses, which often leads to poor financial planning. A simple and efficient system is required to monitor financial transactions and maintain a proper record.

The Personal Finance Management System (Expense Tracker) is a web-based application designed to help users manage their finances effectively. It allows users to record income and expenses, calculate the total balance, and visualize financial data through charts. The system is developed using HTML, CSS, and JavaScript, making it lightweight and easy to use without requiring any backend or database.

1.2 Objectives

The main objectives of this project are:

- To develop a user-friendly system for tracking income and expenses
- To provide real-time calculation of total balance
- To store transaction data using localStorage
- To display monthly financial summary using charts
- To help users understand their spending habits

1.3 Scope of the Project

The scope of the Personal Finance Management System is to provide a simple and efficient way for users to track their daily financial activities. The system is designed primarily for individuals who want to manage their income and expenses without using complex financial software.

This application can be used by students, working professionals, and small households to maintain a record of their financial transactions. It helps users understand their spending patterns and make better financial decisions.

Since the system is web-based, it can be accessed easily through any modern web browser without requiring installation. The use of localStorage ensures that data is stored securely within the user's device, making the system lightweight and private.

The project can also be extended in the future to support advanced features such as cloud storage, user authentication, and multi-device access.

1.4 Problem Statement

Managing personal finances is a challenging task for many individuals. Most people do not maintain a proper record of their daily expenses and income, which often leads to poor financial planning and unnecessary spending.

Traditional methods such as writing expenses in notebooks are time-consuming and inefficient. Additionally, existing digital applications are often complex and include features that are not required by basic users.

Therefore, there is a need for a simple, user-friendly, and efficient system that allows users to track their financial transactions easily. The proposed Personal Finance Management System aims to solve this problem by providing a lightweight and easy-to-use application.

1.5 Applications

The Personal Finance Management System has various practical applications in real life. It can be used by students to track their daily expenses and manage their pocket money effectively.

Working professionals can use the system to monitor their income and expenditures, helping them maintain a balanced financial life. Small households can also use this system to manage their monthly budgets.

Additionally, the system can be used as a learning tool for understanding basic financial management concepts and the use of web technologies in real-world applications.

CHAPTER 2: LITERATURE SURVEY

2.1 Existing Systems

There are many financial tracking applications available such as mobile wallet apps and budgeting tools. These systems provide advanced features like bank integration, automatic transaction tracking, and financial analysis.

2.2 Limitations of Existing Systems

- Despite their features, existing systems have certain limitations:
- Many applications are complex and difficult to use
- Some require internet connectivity at all times
- Data privacy concerns due to online storage
- Not suitable for basic users who need simple tracking

2.3 Need for Proposed System

A simple and lightweight solution is required for users who want to track their finances without complexity. The proposed system provides an easy-to-use interface and stores data locally, ensuring simplicity and privacy.

CHAPTER 3: PROPOSED SYSTEM

3.1 Overview

The proposed system is a web-based Expense Tracker that allows users to manage their income and expenses efficiently. The system provides real-time updates and graphical representation of financial data.

3.2 Working of the System

1. The system works as follows:
2. The user enters transaction details (description and amount)
3. Positive values are treated as income, while negative values are treated as expenses
4. The system stores data using localStorage
5. The balance is updated automatically
6. A monthly chart is generated to show income vs expense

3.3 System Architecture

- The system consists of three main components:
- Frontend (HTML & CSS): Provides user interface
- Logic (JavaScript): Handles calculations and data processing
- Storage (localStorage): Stores user data in browser

3.4 Advantages of Proposed System

- Simple and easy to use
- No internet required after loading
- Real-time updates
- Lightweight and fast
- Provides visual representation using charts

3.5 Modules of the System

The system is divided into different modules to ensure smooth functioning and better organization.

The Input Module allows users to enter transaction details such as description, amount, and category.

The Processing Module handles the logic of calculating total balance, income, and expenses based on the entered data.

The Storage Module uses localStorage to store user data in the browser, ensuring data persistence.

The Visualization Module displays financial data using charts, helping users understand their financial status easily.

3.6 Algorithm

The working of the system can be explained using the following algorithm:

Step 1: Start the system

Step 2: Accept transaction details from the user

Step 3: Validate the input data

Step 4: Store the transaction in localStorage

Step 5: Update total balance, income, and expenses

Step 6: Display updated data on the screen

Step 7: Generate chart for monthly summary

Step 8: Stop

3.7 Flowchart of the System

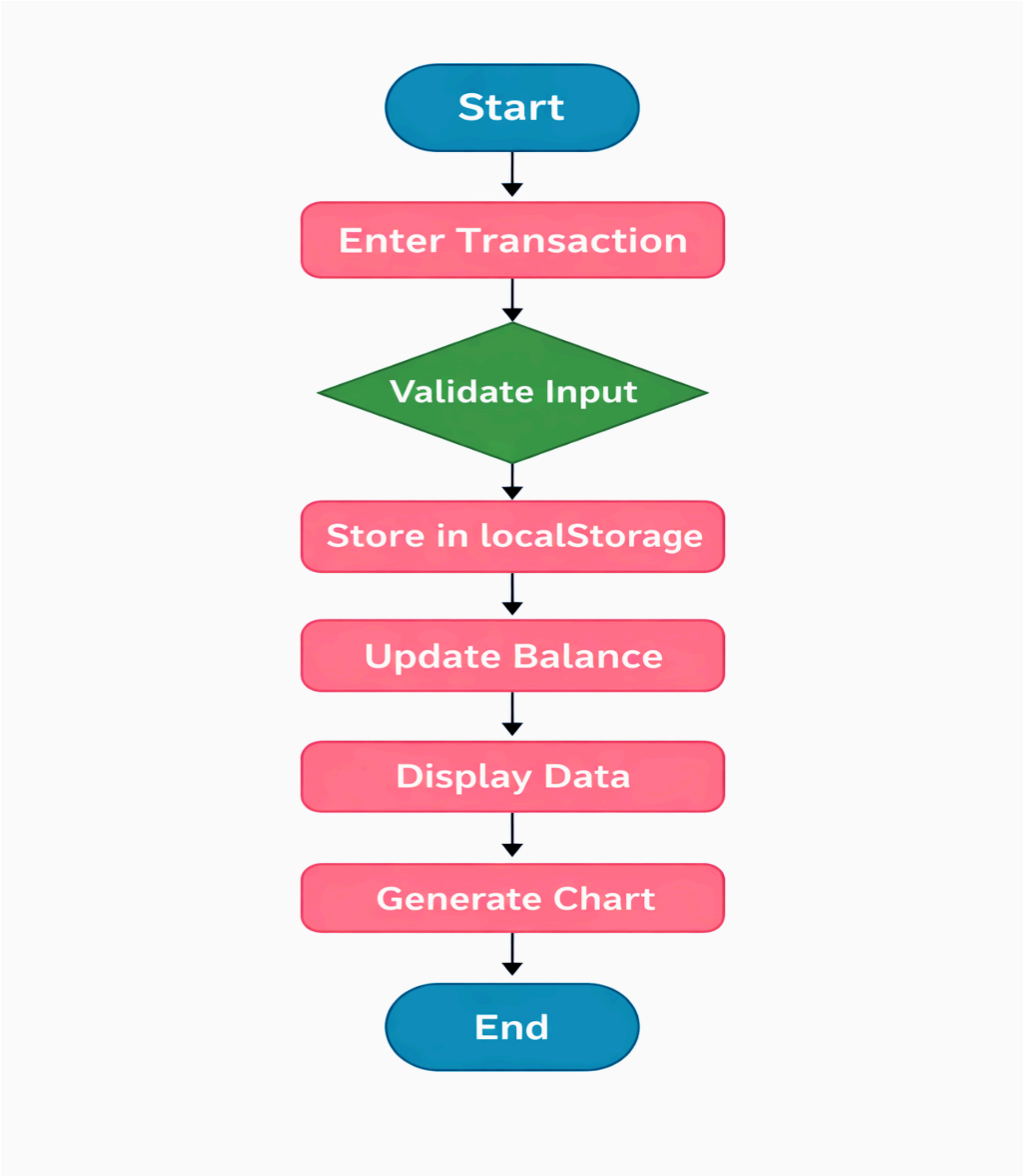
The flowchart represents the step-by-step working process of the Personal Finance Management System. It illustrates how user input is processed and how the system generates output.

The process begins when the user enters transaction details such as description, amount, and category. The system then validates the input to ensure correctness. Once validated, the data is stored in the browser using localStorage.

After storing the data, the system updates the balance, income, and expense values. The updated information is displayed to the user along with a list of transactions. Finally, a chart is generated to visually represent the monthly summary of income and expenses.

This flowchart helps in understanding the internal working of the system in a clear and structured manner.

- Diagram



3.8 System Architecture

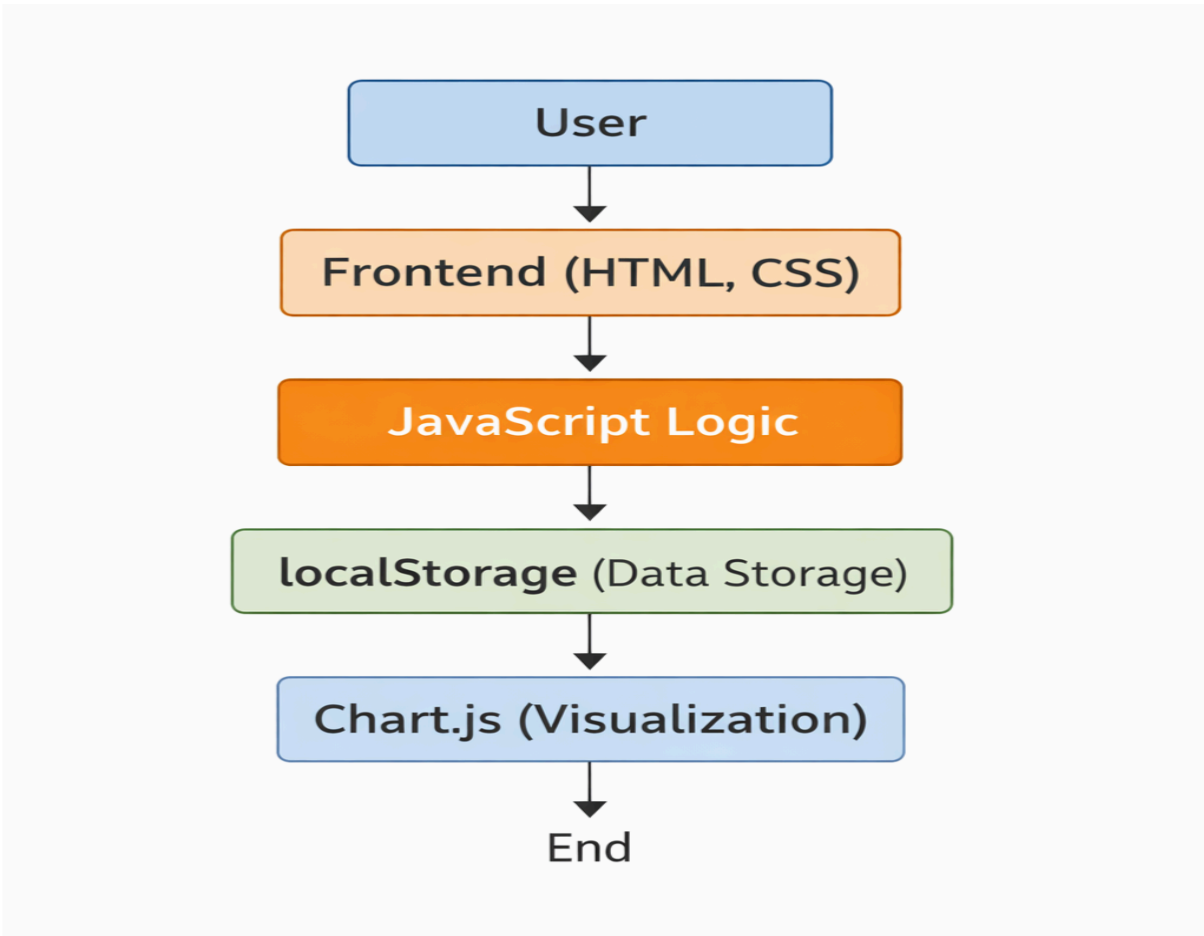
The system architecture of the Personal Finance Management System follows a simple client-side model. It consists of multiple components that work together to provide a smooth and efficient user experience.

The User interacts with the system through the web interface. The Frontend layer, developed using HTML and CSS, is responsible for displaying the user interface.

The JavaScript layer handles all the logic and processing of data, including adding transactions, calculating balance, and updating the interface dynamically.

The data is stored locally in the browser using localStorage, which ensures data persistence without requiring a backend server. For visualization purposes, Chart.js is used to generate graphical representations of income and expense data.

This architecture makes the system lightweight, fast, and easy to maintain.



CHAPTER 4: RESULTS AND DISCUSSION

4.1 Results

The developed system successfully performs all the required operations:

- Users can add and delete transactions
- Balance is calculated automatically
- Income and expense are displayed separately
- Monthly chart shows financial summary
- Data is stored and retrieved using localStorage

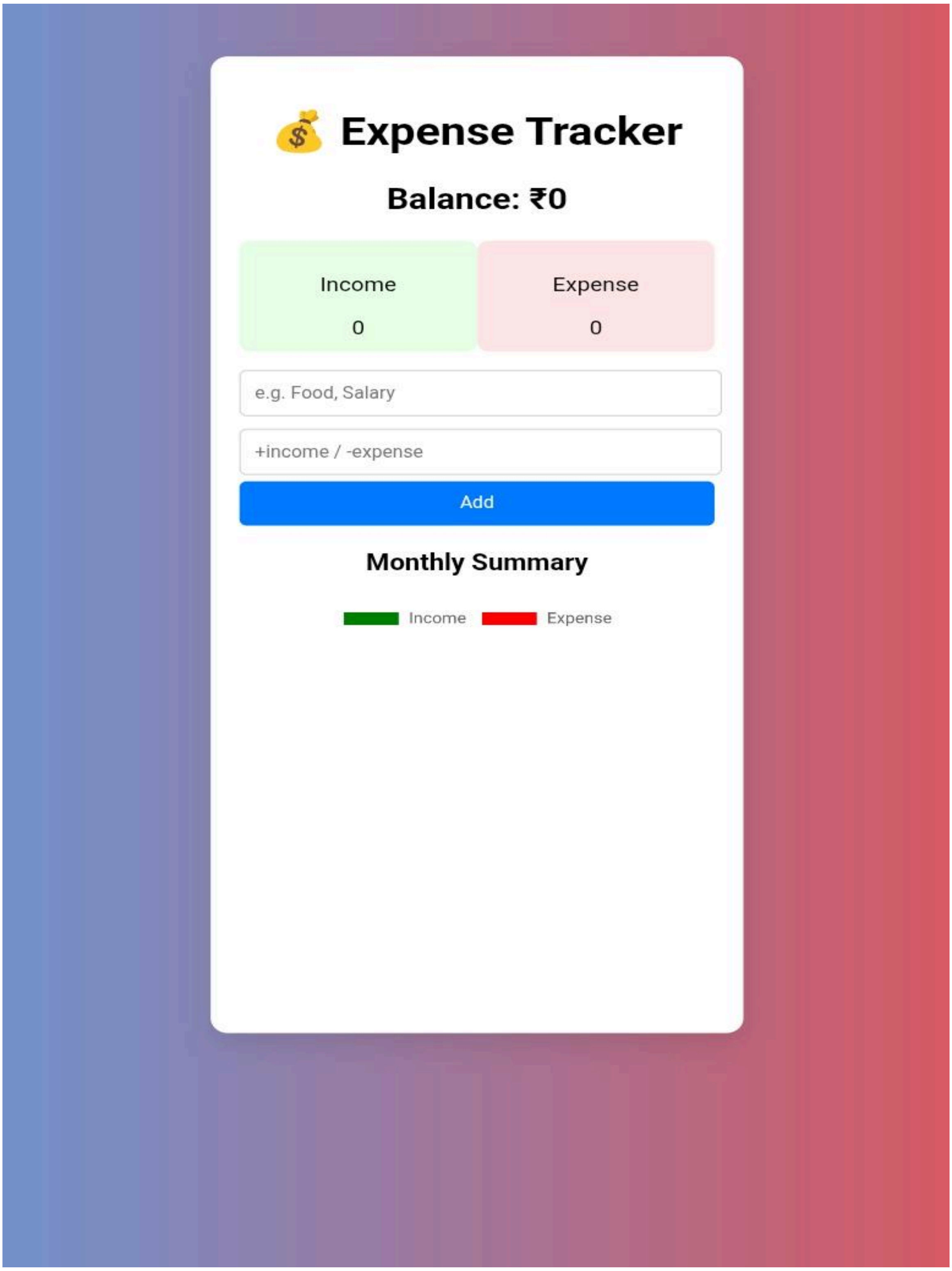
4.2 Discussion

The results show that the system is effective in managing personal financial data in a simple and user-friendly manner. The use of localStorage ensures that data is retained even after refreshing the page. Features such as filtering transactions and category-based organization help users analyze their spending patterns more clearly.

The integration of chart visualization provides a graphical representation of monthly income and expenses, making the system more interactive and informative. Overall, the system meets its objectives and provides a reliable solution for basic financial management. However, the absence of advanced features such as user authentication and cloud storage limits its scalability for larger applications..

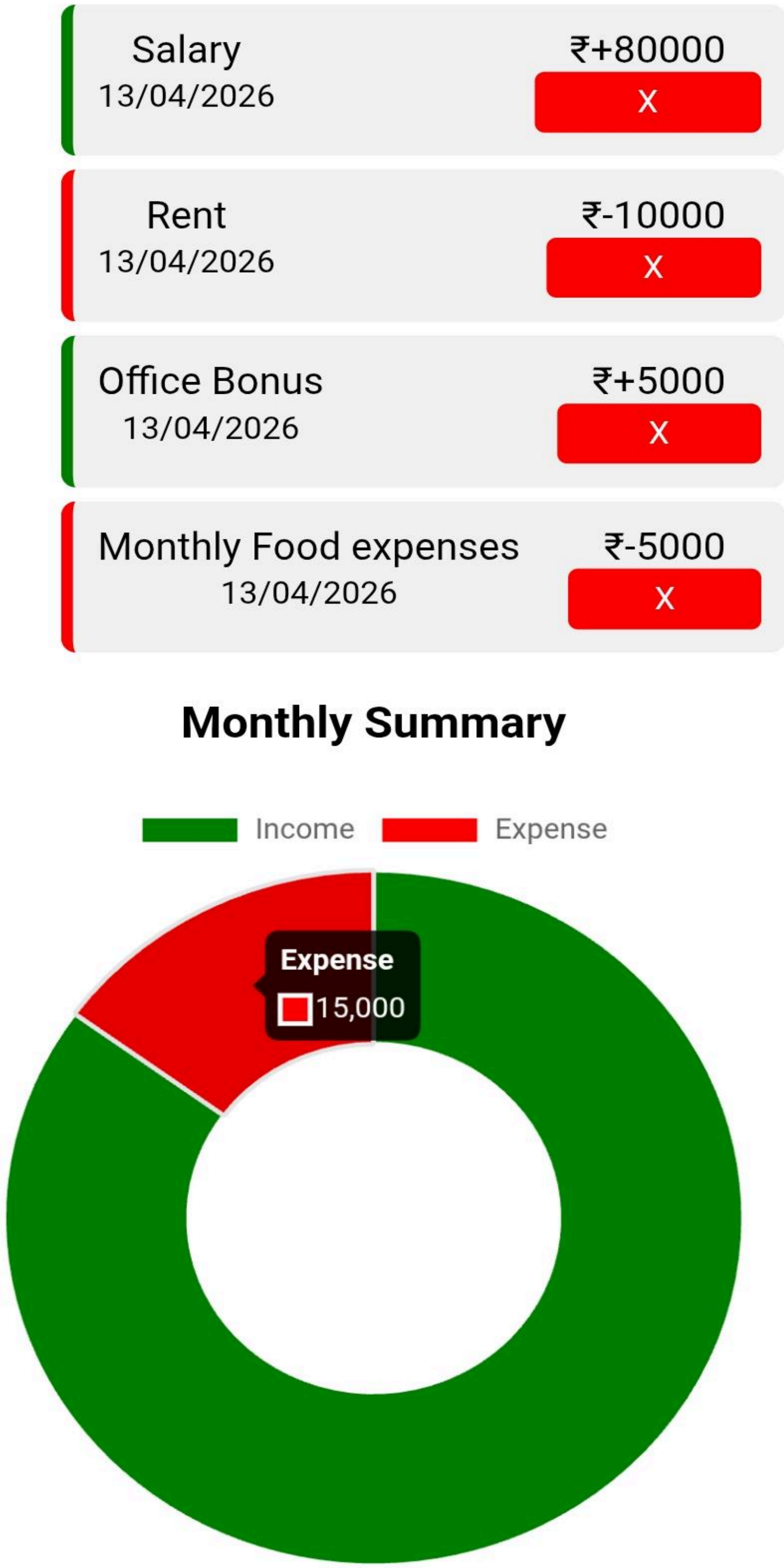
4.3 Screenshots

1. Main interface

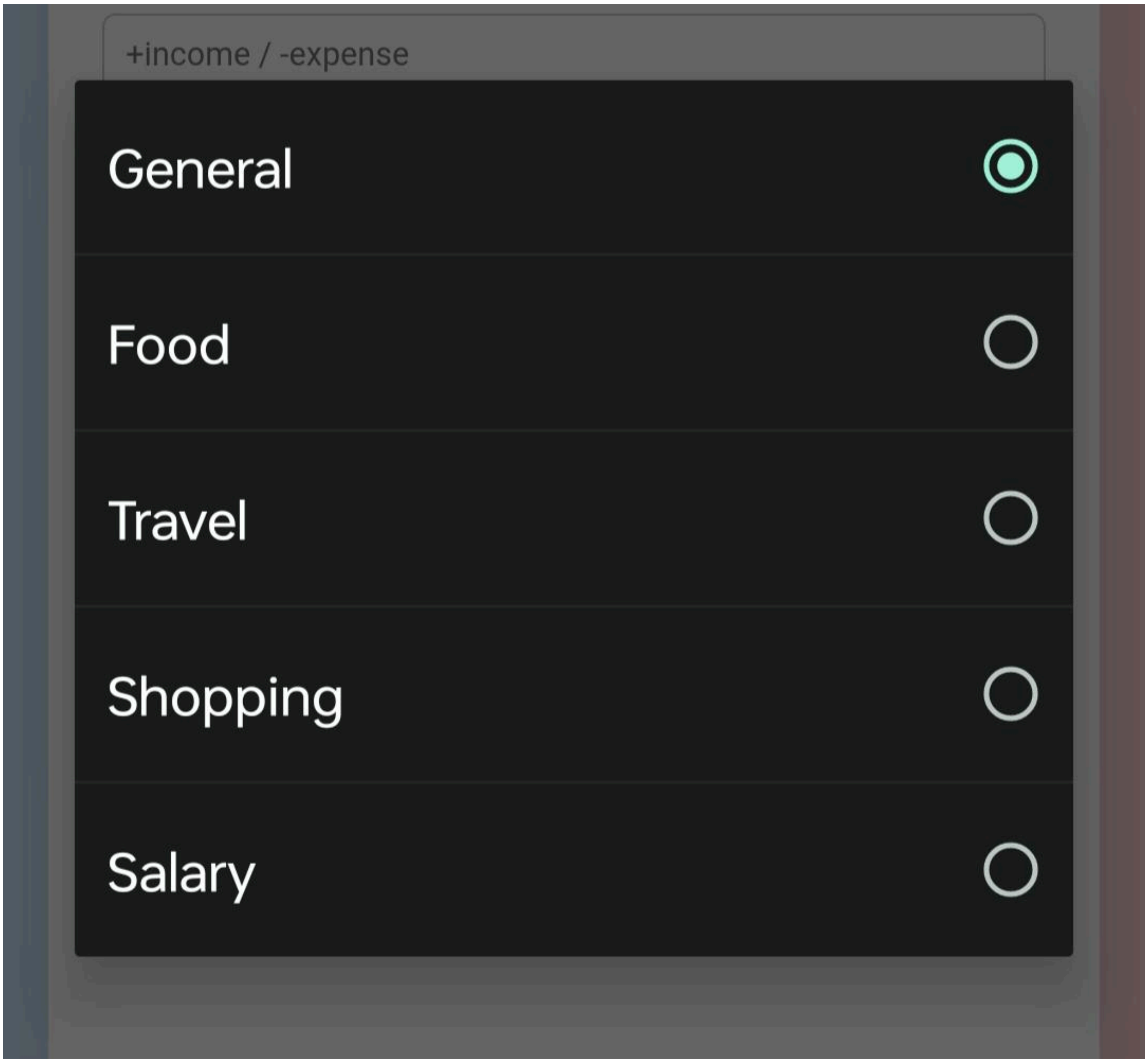


2. Transaction list

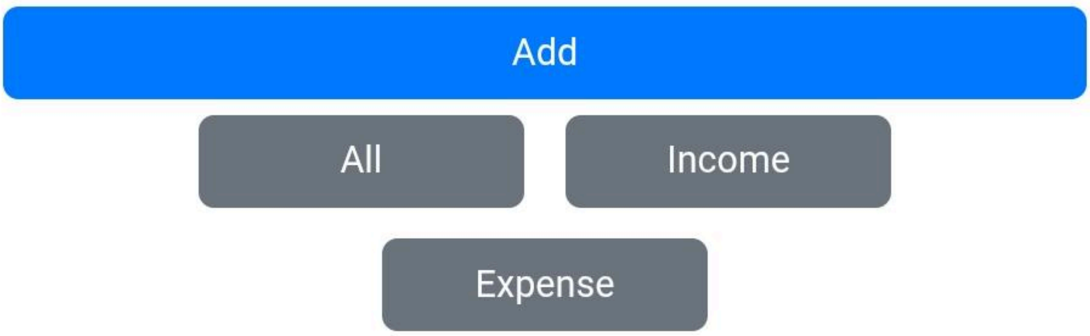
3. Chart View


Salary
13/04/2026
₹+80000
X
Rent
13/04/2026
₹-10000
X
Office Bonus
13/04/2026
₹+5000
X
Monthly Food expenses
13/04/2026
₹-5000
X
Monthly Summary
Income
Expense
Expense
15,000

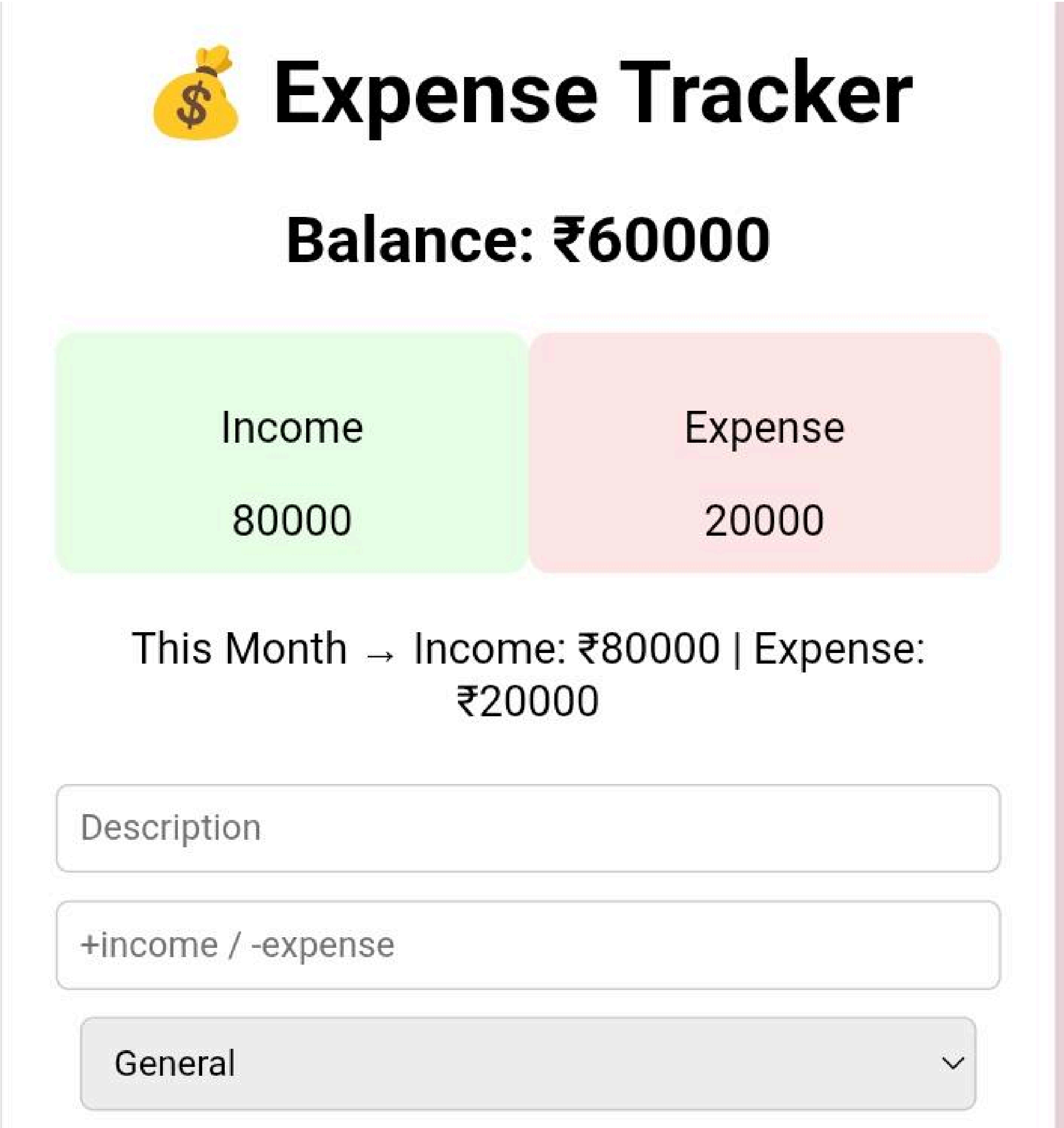
4. Category-based Transactions



5. Filtering Transactions



6. Monthly Summary Display



7. HTML Transaction code

```html

<div class="form">
  <input type="text" id="desc" placeholder="e.g. Food, Salary">
  <input type="number" id="amount" placeholder="+income / -expense">
  <button id="addBtn">Add</button>
</div>

```

8. JavaScript Important Logic

```
expenses tracker ~zq88bir6...    index.html    script.js

function addTransaction() {
  const desc = descInput.value.trim();
  const amount = parseFloat(amountInput.value);

  if (desc === "" || isNaN(amount)) {
    alert("Enter valid data");
    return;
  }

  const transaction = {
    id: Date.now(),
    desc,
    amount,
    date: new Date().toISOString() // important for chart
  };

  transactions.push(transaction);
  updateLocalStorage();
  render();

```

9. Chart (JavaScript)

```
    chart = new Chart(ctx, {
      type: "doughnut",
      data: {
        labels: ["Income", "Expense"],
        datasets: [{
          data: [income, expense],
          backgroundColor: ["green", "red"]
        }]
      }
    });
}
```

CHAPTER 5: TESTING AND VALIDATION

5.1 Introduction

Testing is an essential phase in the development of any software system. It ensures that the application works correctly and meets the required objectives. In the Personal Finance Management System, testing was performed to verify that all features function properly and provide accurate results.

The system was tested under different conditions to identify errors and ensure reliability. Both functional and user-level testing were carried out to evaluate the performance of the application

5.2 Testing Objectives

The main objective of testing is to ensure that the system performs all operations correctly. It also aims to identify any errors or bugs present in the system.

Testing helps in verifying that the system meets user requirements and provides accurate output. It also ensures that the system is stable and can handle different types of user inputs without failure.

5.3 Types of Testing

Different types of testing were performed on the system. Functional testing was carried out to check whether all features such as adding, deleting, and filtering transactions work correctly.

User interface testing was performed to ensure that the system is easy to use and visually clear. Data validation testing was also conducted to verify that incorrect or empty inputs are handled properly by the system.

5.4 Test Cases

Several test cases were designed to validate the functionality of the system. These include adding a valid transaction, deleting a transaction, filtering transactions, and checking balance updates.

Each test case was executed and the results were compared with expected outcomes. The system produced correct results in all cases, confirming its proper functioning.

Test case Tabular form :

Test Case	**Input**	**Expected Output**	**Result**
Add Transaction	₹500	Added to list	Pass
Delete	Remove Item	Item Removed	Pass
Filter	Income	Show only income	Pass

Features-wise Testing :

Feature	**Description**	**Expected outcome**	**Result**
Add Transaction	User enters amount and description	Transaction added successfully	Pass
Delete Transaction	User clicks delete button	Transaction removed from list	Pass
Category selection	User selects category	Category displayed with transaction	Pass
Filter system	User selects income/expense filter	Only selected type is displayed	Pass
Balance calculation	System calculates Total Balance	Correct balance is shown	Pass
Local storage	Data stored in browser	Data persists after refresh	Pass
Chart visualization	System generates chart	Chart displays income vs expense correctly	Pass

5.5 System Limitations

Although the Personal Finance Management System performs its intended functions effectively, it has certain limitations. The system does not include user authentication, which means that there is no login or account-based security for users.

The application stores data using localStorage, which restricts access to a single device and browser. This limits the portability of data and prevents users from accessing their financial records across multiple devices.

Additionally, the system does not provide advanced financial analysis features such as budgeting, forecasting, or automated insights. These limitations highlight areas where the system can be improved in future versions.

5.6 Future Enhancements

The Personal Finance Management System can be enhanced further by adding advanced features to improve its functionality and usability. One major improvement would be the integration of user authentication, allowing users to create accounts and securely store their data.

The system can also be upgraded to use cloud-based storage instead of localStorage, enabling users to access their data from multiple devices. Additional features such as budget planning, expense reminders, and financial analysis tools can also be implemented.

Furthermore, the application can be converted into a mobile app to increase accessibility. These enhancements would make the system more powerful and suitable for real-world applications.

CHAPTER 6: CONCLUSION & FUTURE WORK

6.1 Conclusion

The Personal Finance Management System was successfully designed and implemented to help users manage their daily financial activities in a simple and efficient manner. The system allows users to add, delete, and categorize transactions while providing real-time updates of balance, income, and expenses.

The project demonstrates the effective use of web technologies such as HTML, CSS, and JavaScript in developing a practical application. The integration of features like filtering, category management, and graphical representation using charts enhances the usability and functionality of the system.

The application provides a clear understanding of spending patterns and helps users make better financial decisions. Its user-friendly interface ensures that even non-technical users can easily interact with the system without any difficulty.

Overall, the system meets its objectives and serves as a reliable tool for basic financial management. It highlights the importance of simple digital solutions in improving everyday tasks and demonstrates the potential of web-based applications in solving real-world problems.

6.2 Future Work

Although the Personal Finance Management System performs its intended functions effectively, there is significant scope for further improvement. One of the major enhancements would be the addition of user authentication, allowing users to securely store and manage their financial data through login credentials.

The system can be improved by integrating cloud-based storage, which would enable users to access their data from multiple devices and ensure better data security. Advanced features such as budget planning, expense tracking reports, and automated financial insights can also be added to enhance the functionality of the application.

In addition, the system can be extended to include notifications and reminders for expenses, helping users maintain better financial discipline. The development of a mobile application version would further increase accessibility and convenience for users.

Future improvements can also focus on improving the user interface and adding customization options to enhance the overall user experience. With these enhancements, the system can evolve into a more powerful and scalable financial management solution suitable for real-world applications.

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